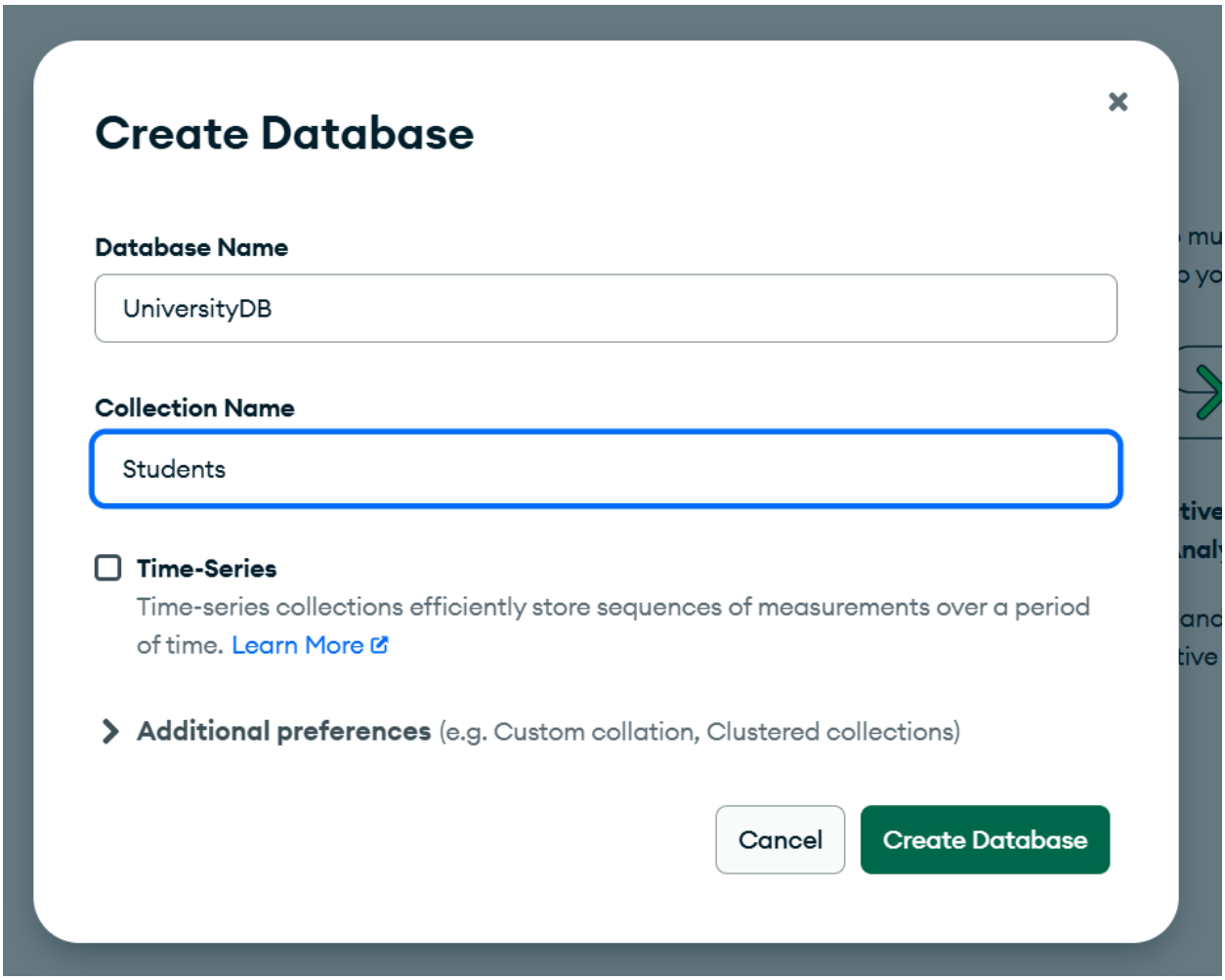
PORTFOLIO 4 – Portfolio 4: NoSQL and Advanced Data Models

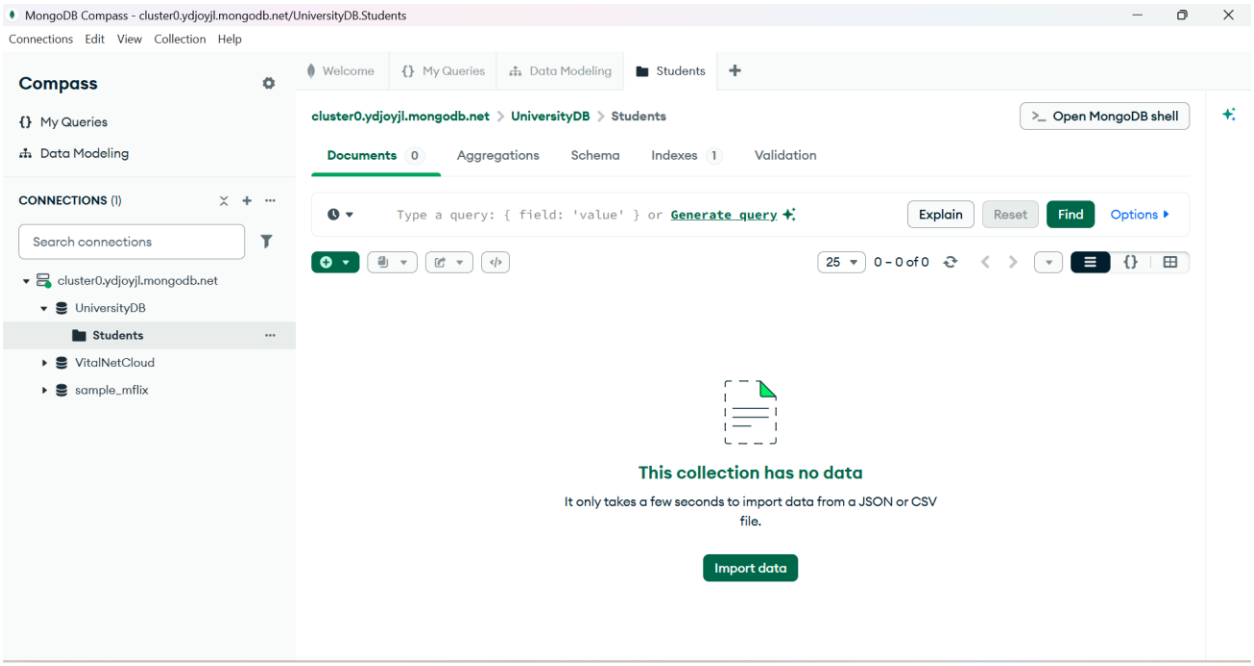
I began by downloading **MongoDB Compass** to use to Create and Visualize my Database. I then connected it to my Miva **MongoDB Atlas** by Permitting my IP Address.

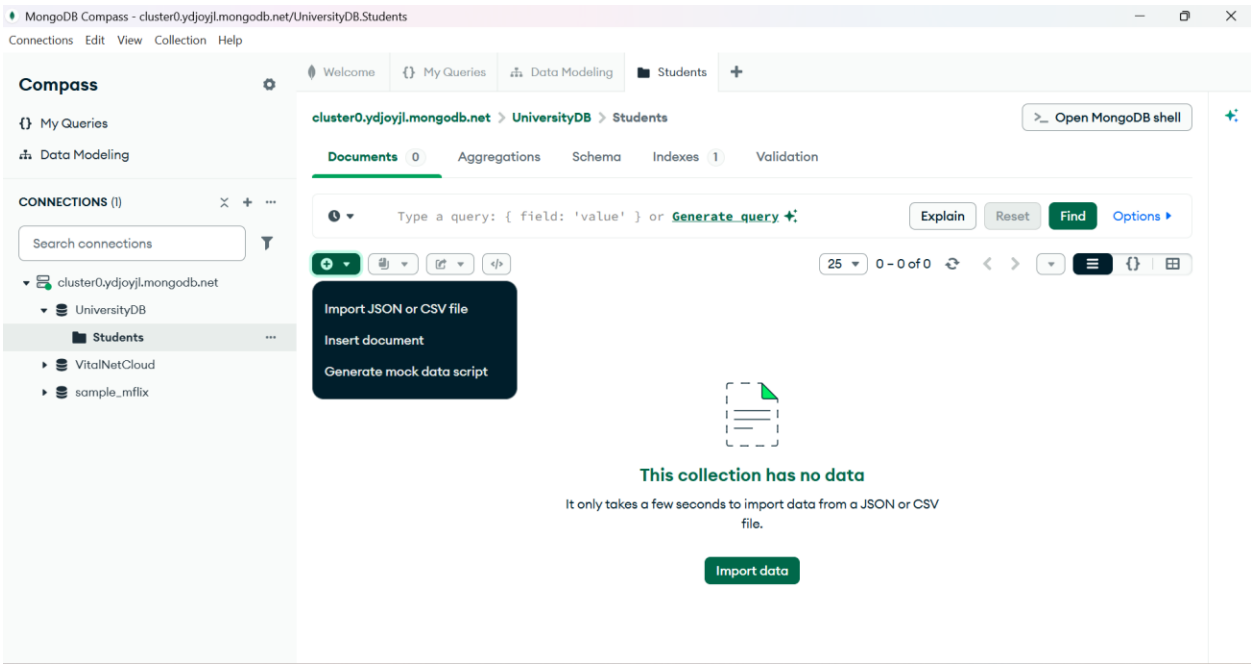
After a successful connection, I performed:

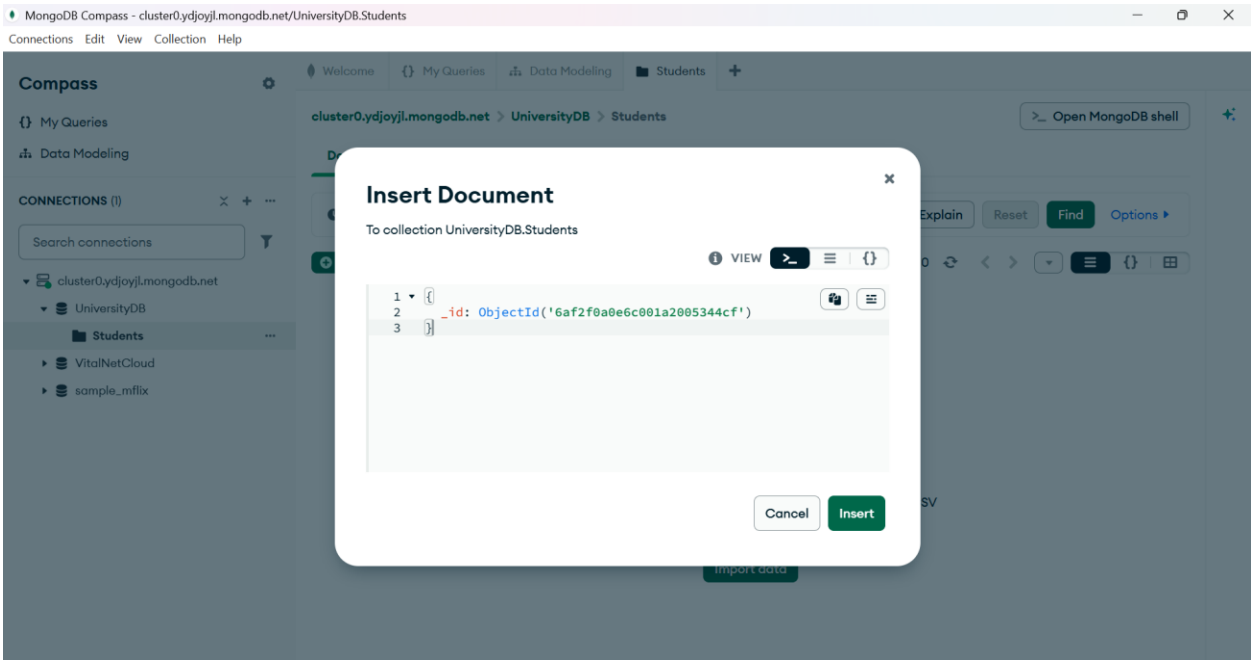
(1) Database Creation
(2) Collection Creation
(3) Document Insertion
(4) Document Retrieval



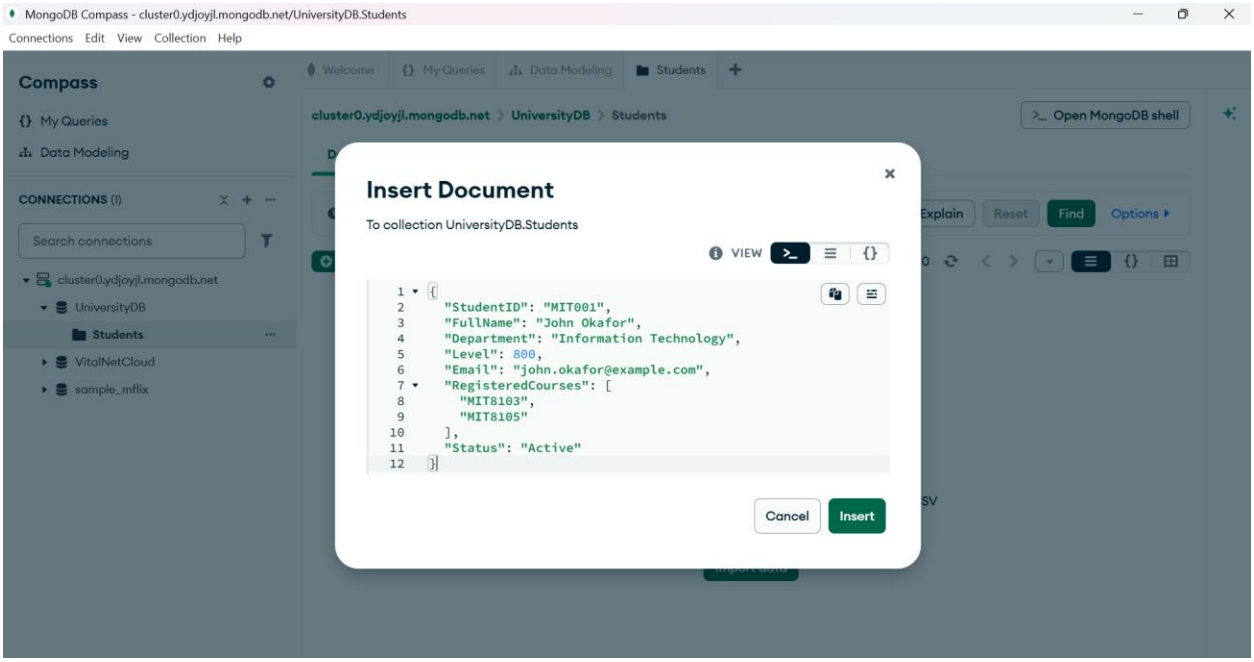
After creating the Database, below is the screenshot.

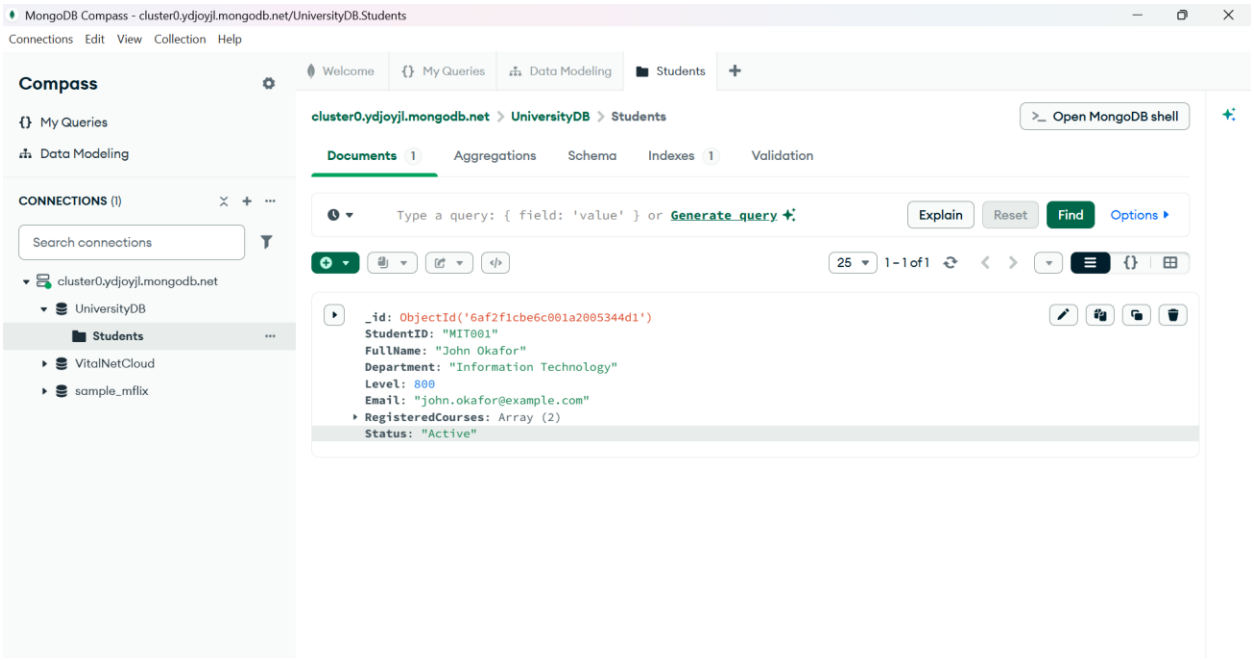




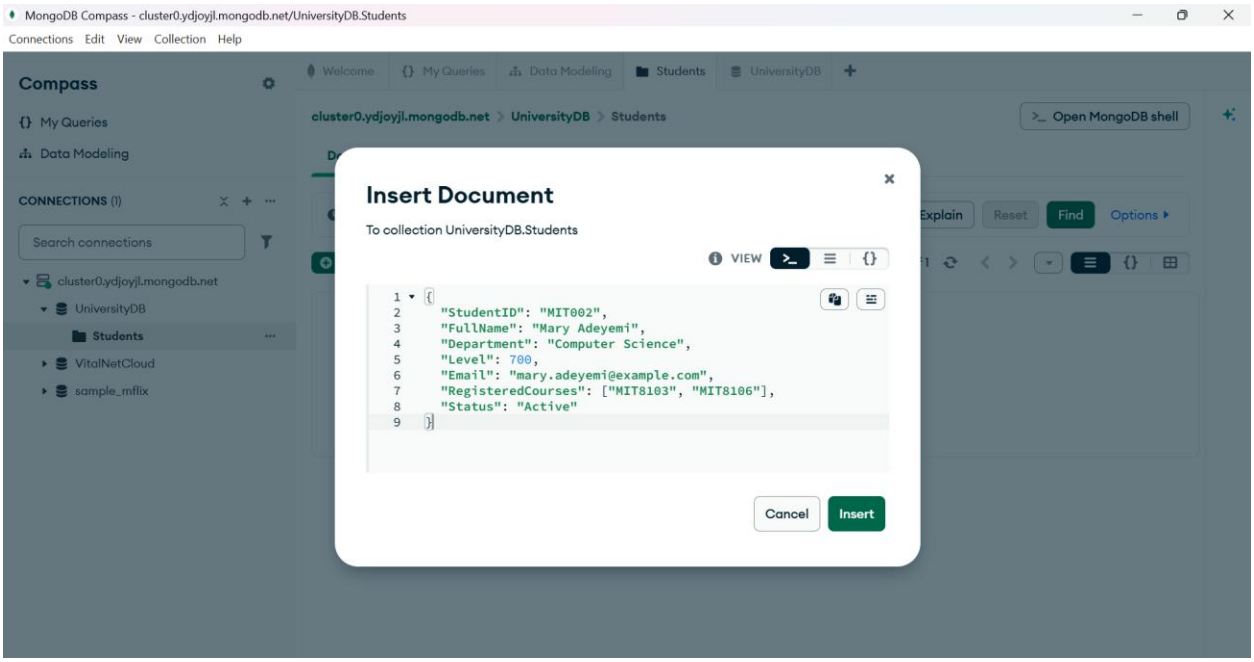


The Student **John Okafor** was created.

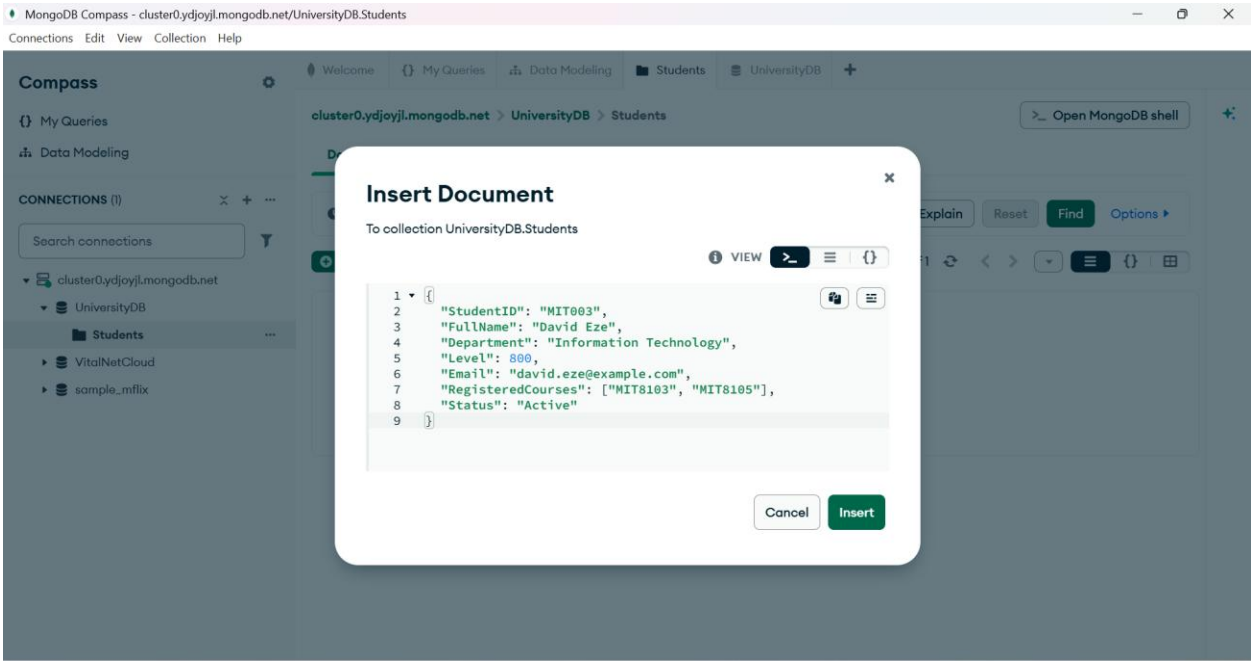




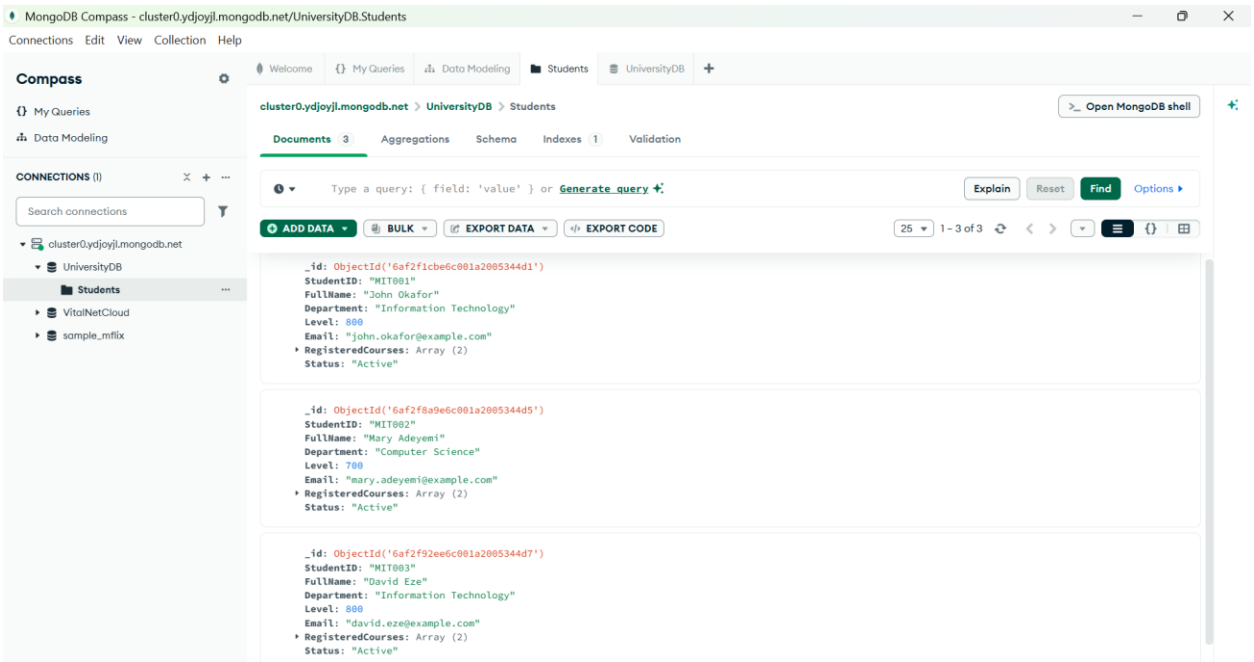
The Student **Mary Adeyemi** was created



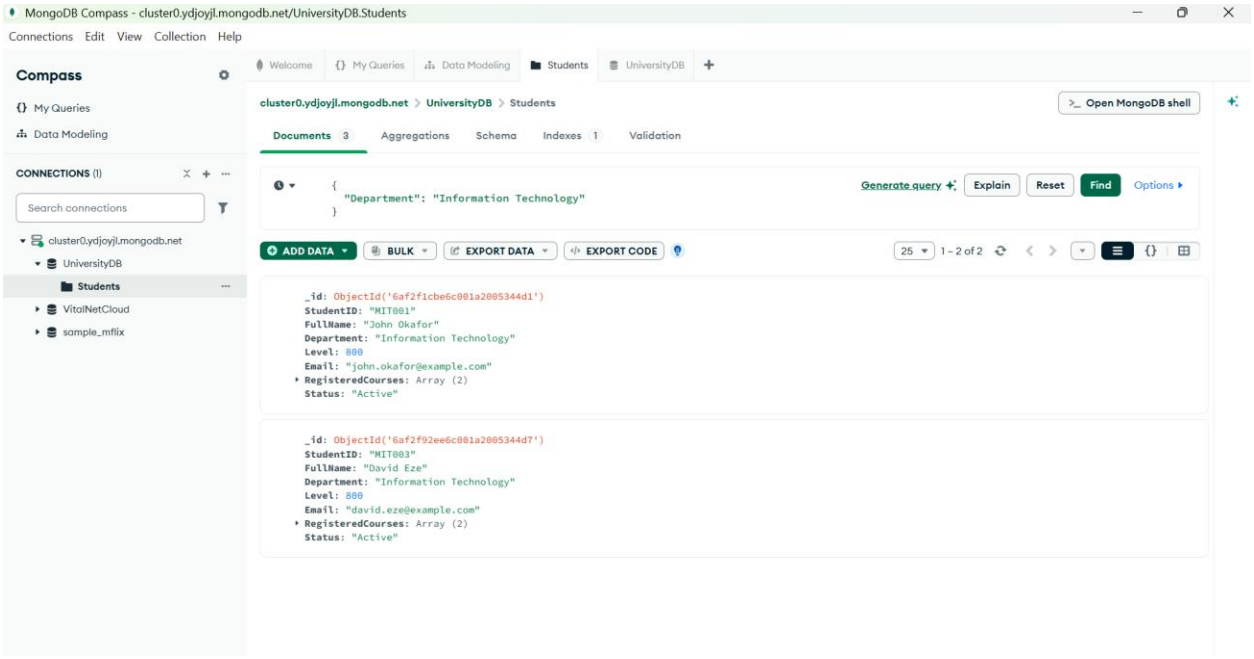
The Student **David Eze** was created.



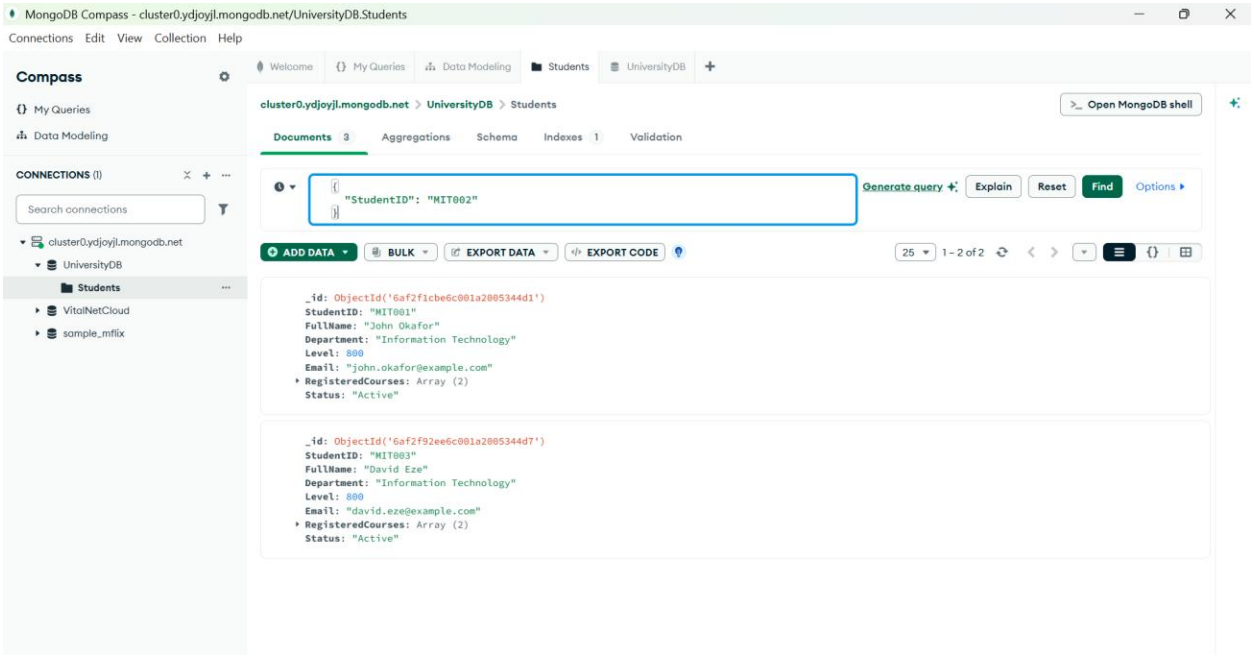
These are the **three Student Records.**

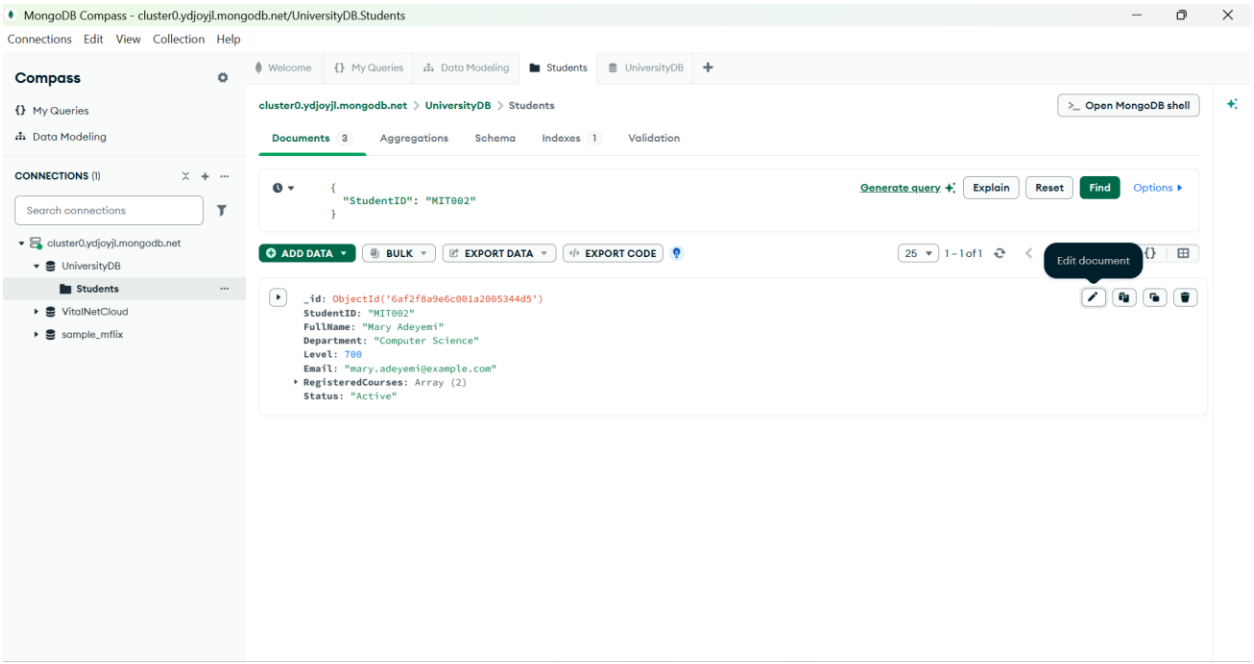


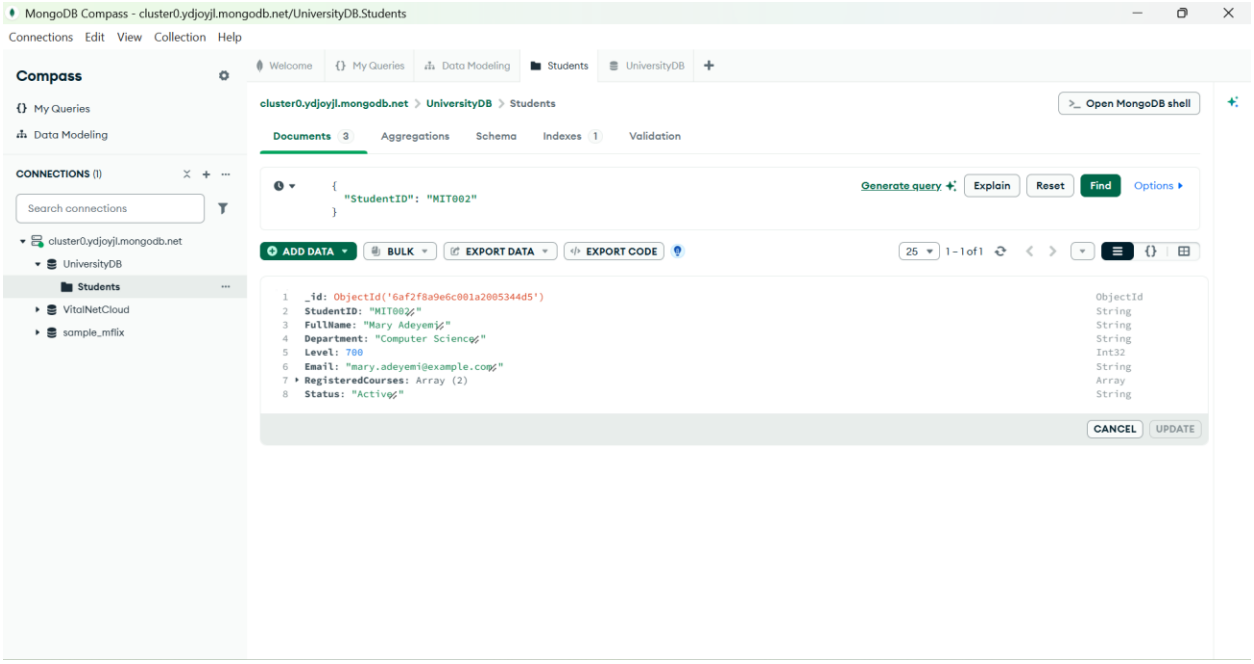
READ OPERATION: To find students in the Information Technology Department

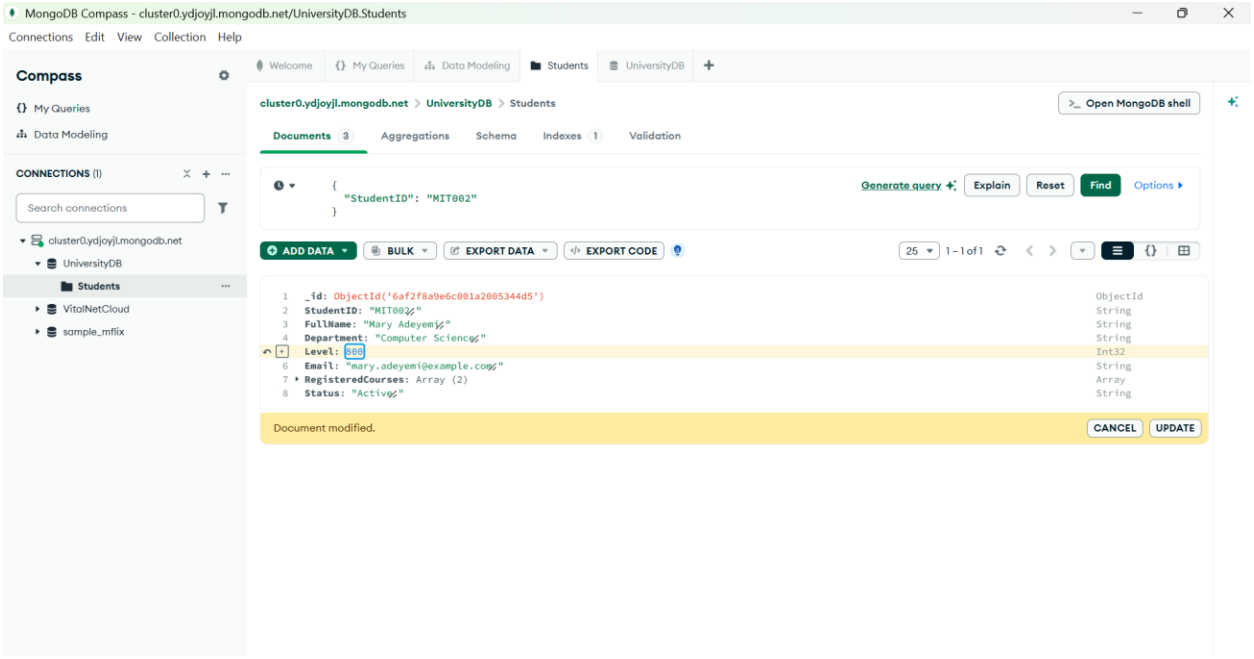


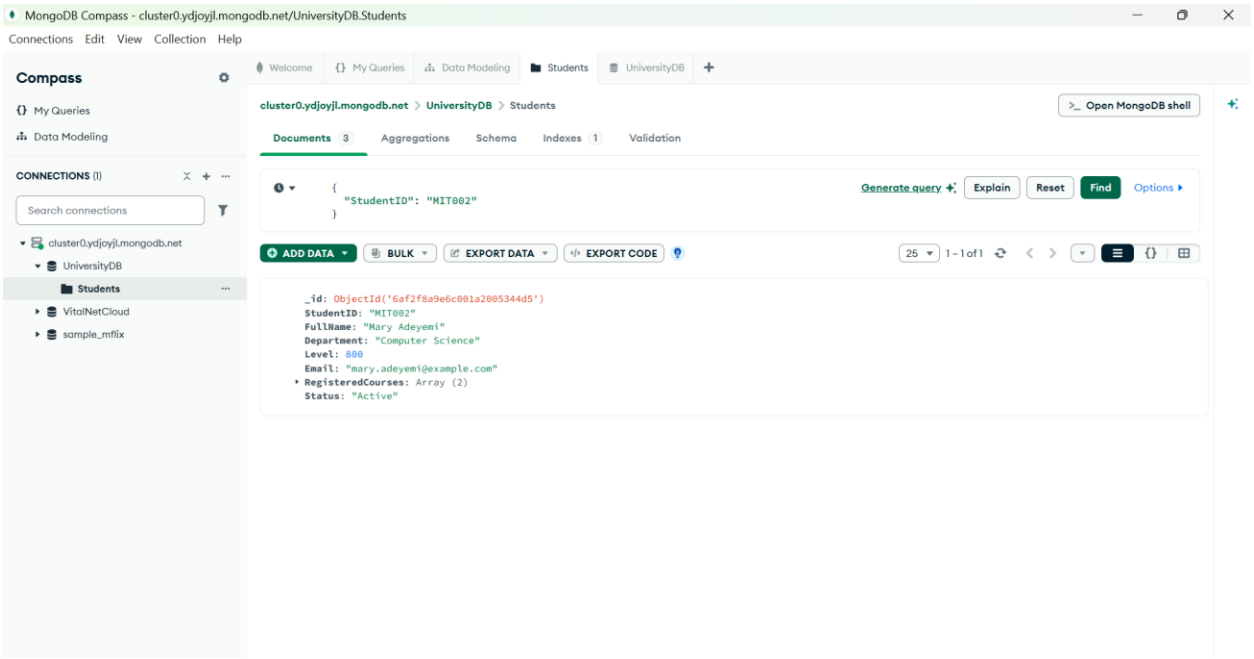
UPDATE OPERATION: Changing Mary Adeyemi's Level to 800



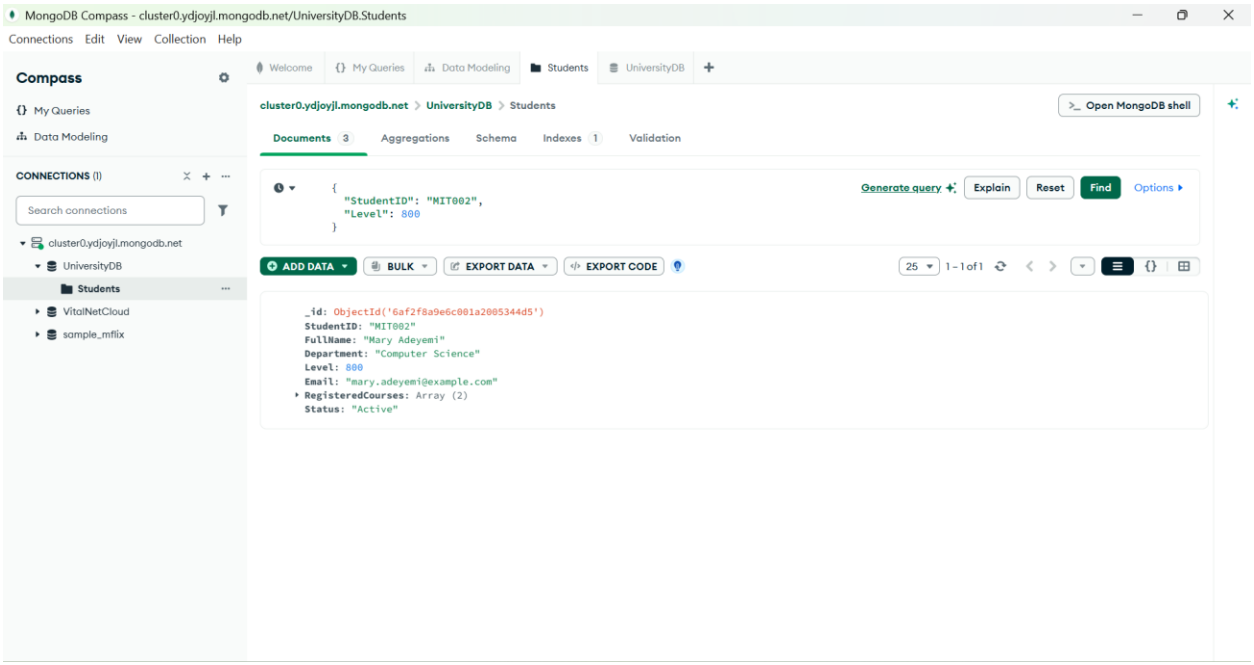

MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{
"StudentID": "MIT002"
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
Edit document
_id: ObjectId('6af2f8a9e6c001a2005344d5')
StudentID: "MIT002"
FullName: "Mary Adeyemi"
Department: "Computer Science"
Level: 700
Email: "mary.adeyemi@example.com"
RegisteredCourses: Array (2)
Status: "Active"


MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{
"StudentID": "MIT002"
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
1 _id: ObjectId('6af2f8a9e6c001a2005344d5') ObjectId
2 StudentID: "MIT002" String
3 FullName: "Mary Adeyemi" String
4 Department: "Computer Science" String
5 Level: 700 Int32
6 Email: "mary.adeyemi@example.com" String
7 RegisteredCourses: Array (2) Array
8 Status: "Active" String
CANCEL
UPDATE

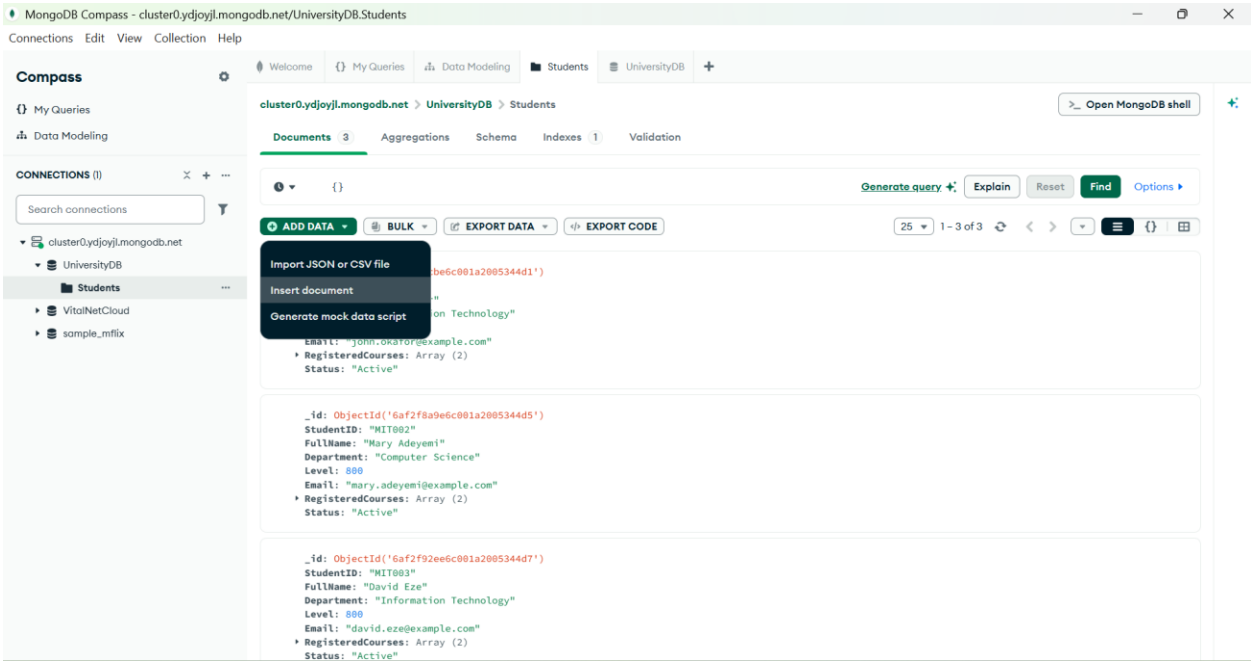

MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{ "StudentID": "MIT002" }
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('6af2f8a9e6c001a2005344d5')
StudentID: "MIT002"
FullName: "Mary Adeyemi"
Department: "Computer Science"
Level: 800
Email: "mary.adeyemi@example.com"
RegisteredCourses: Array (2)
Status: "Active"
ObjectId
String
String
String
Int32
String
Array
String
Document modified.
CANCEL
UPDATE

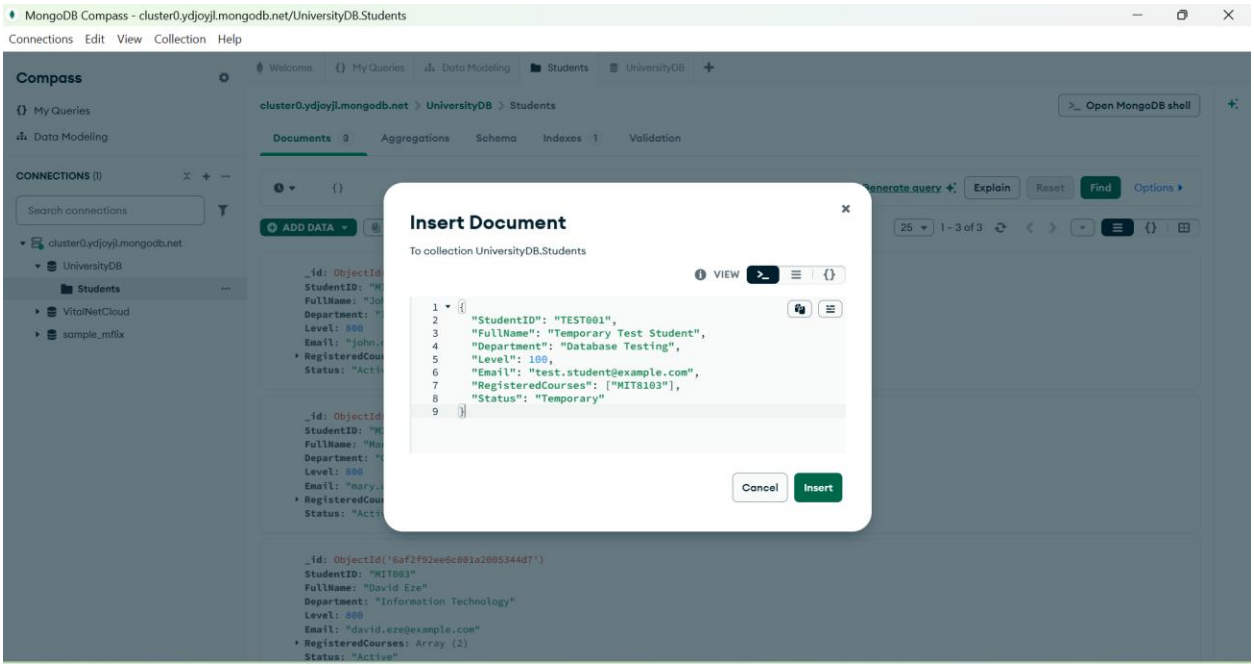

MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{ "StudentID": "MIT002" }
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('6af2f8a9e6c001a2005344d5')
StudentID: "MIT002"
FullName: "Mary Adeyemi"
Department: "Computer Science"
Level: 800
Email: "mary.adeyemi@example.com"
RegisteredCourses: Array (2)
Status: "Active"

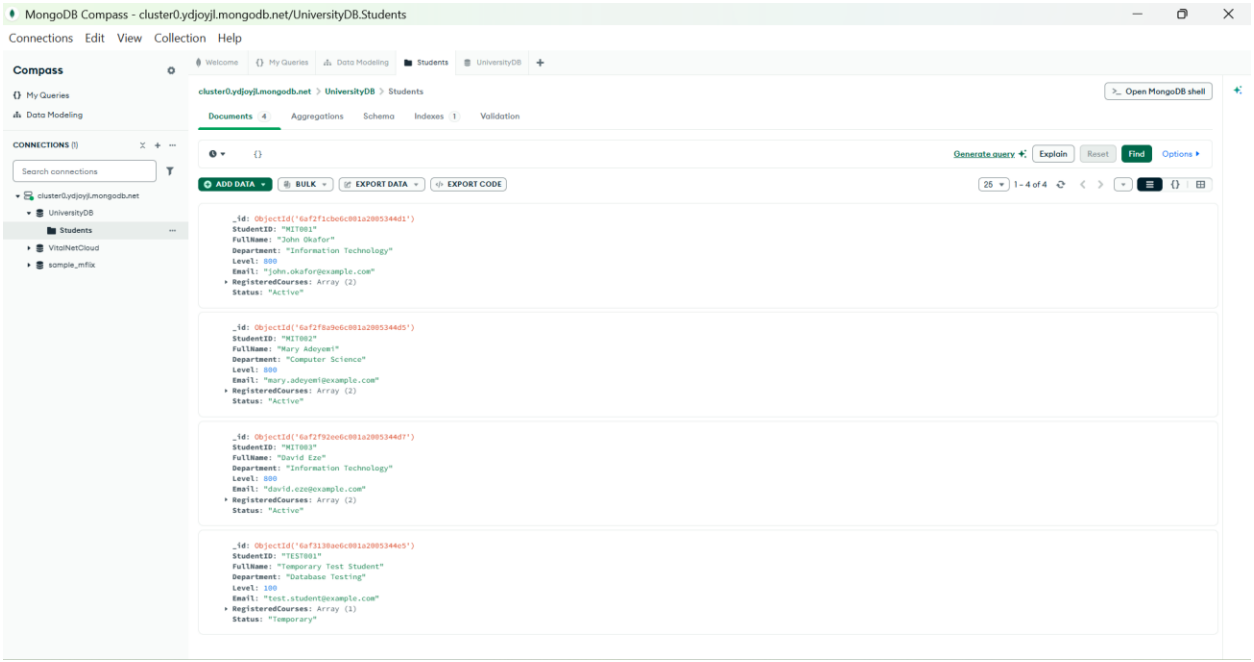
Verifying the Update.

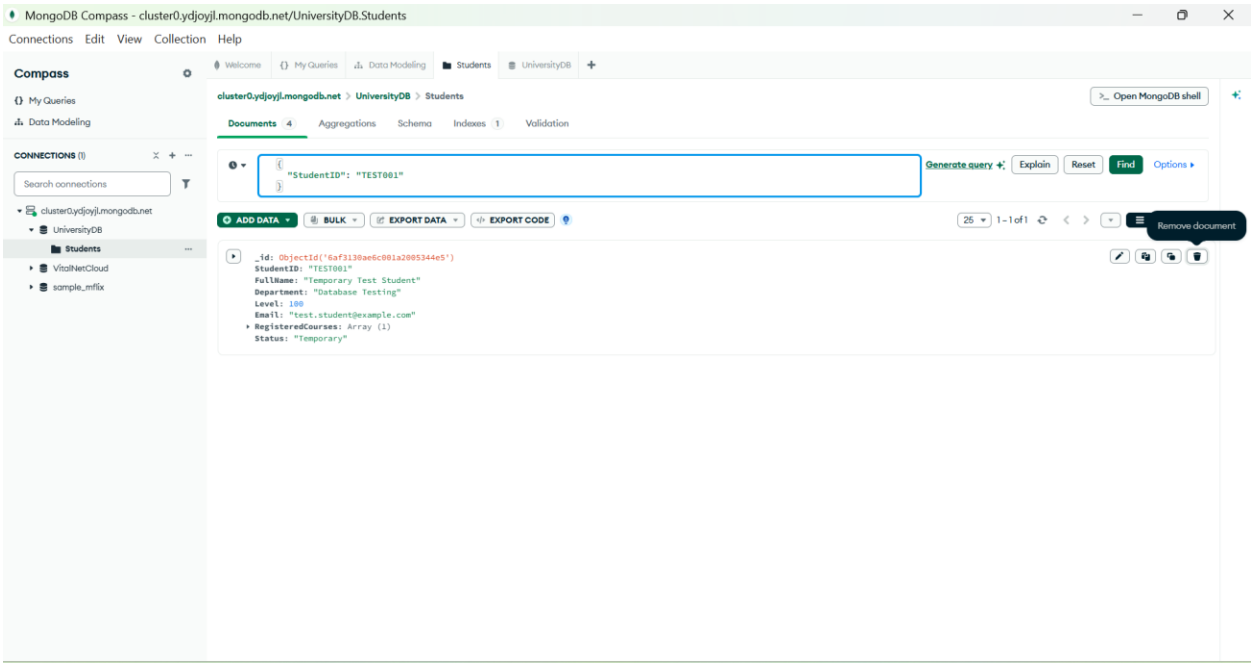


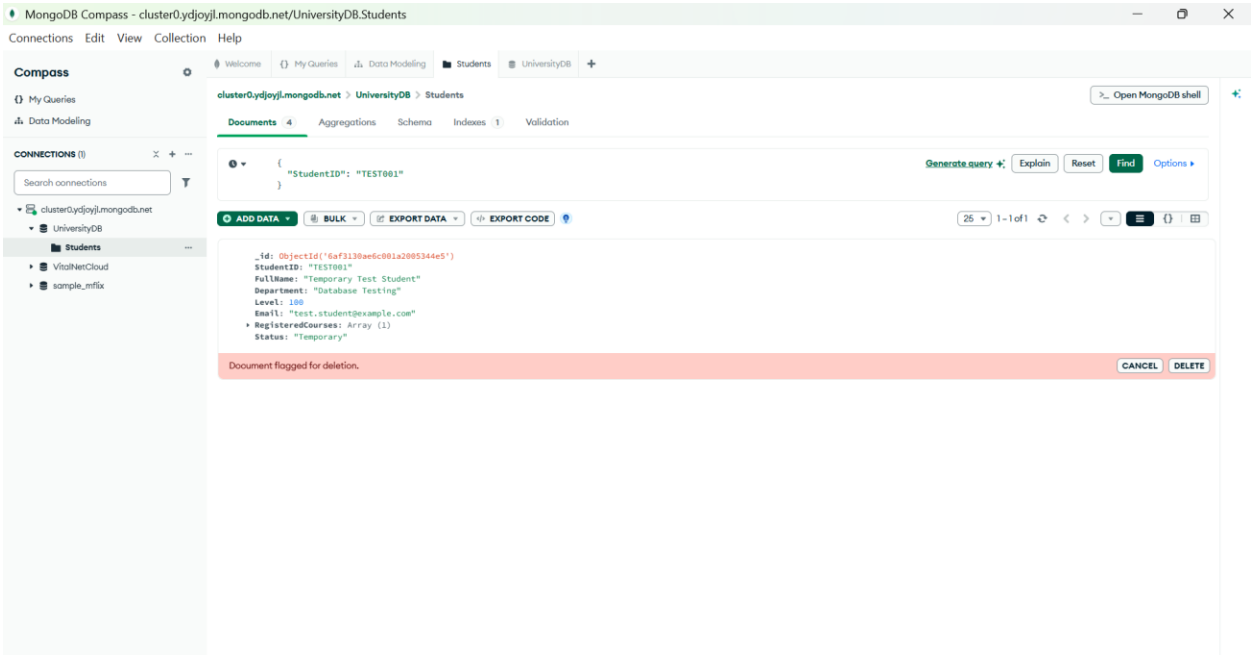
DELETE OPERATION: To delete a Student Record (I added a test student and deleted it)

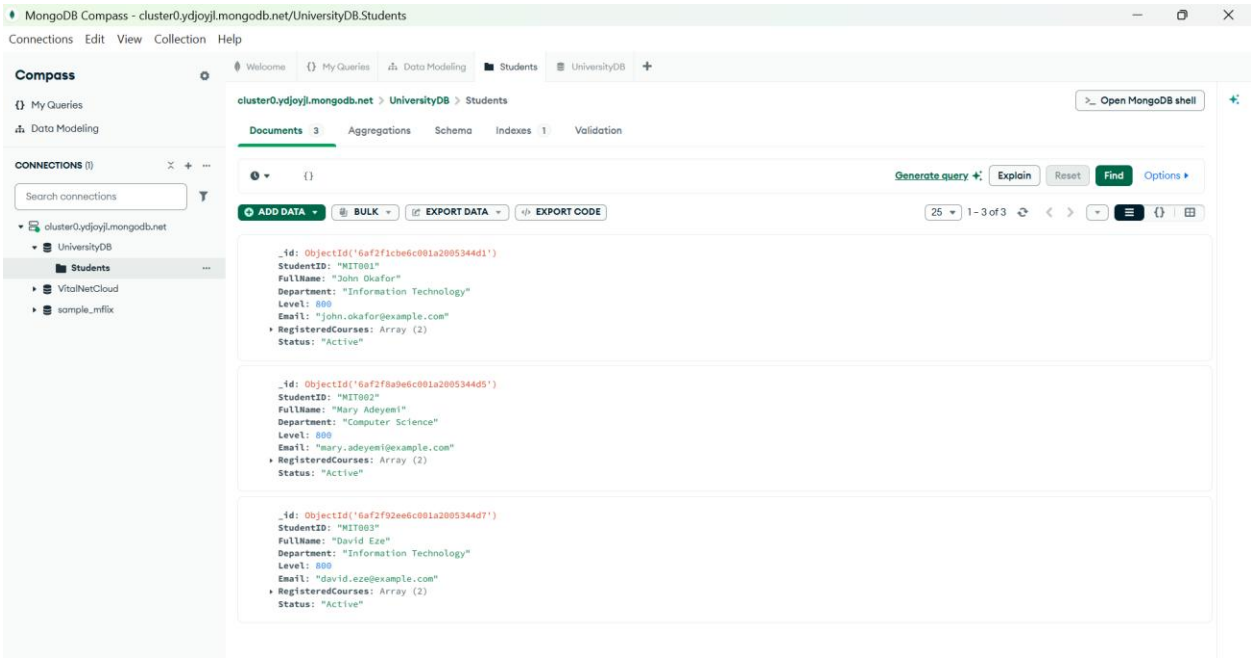



MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
Generate query
Explain
Reset
Find
Options
ADD DATA
1 - 3 of 3
Insert Document
To collection UniversityDB.Students
VIEW
{
"StudentID": "TEST001",
"FullName": "Temporary Test Student",
"Department": "Database Testing",
"Level": 100,
"Email": "test.student@example.com",
"RegisteredCourses": ["MIT8103"],
"Status": "Temporary"
}
Cancel
Insert
_id: ObjectId('6af2f92ee6c801a2005344d7')
StudentID: "MIT003"
FullName: "David Eze"
Department: "Information Technology"
Level: 800
Email: "david.eze@example.com"
RegisteredCourses: Array (2)
Status: "Active"

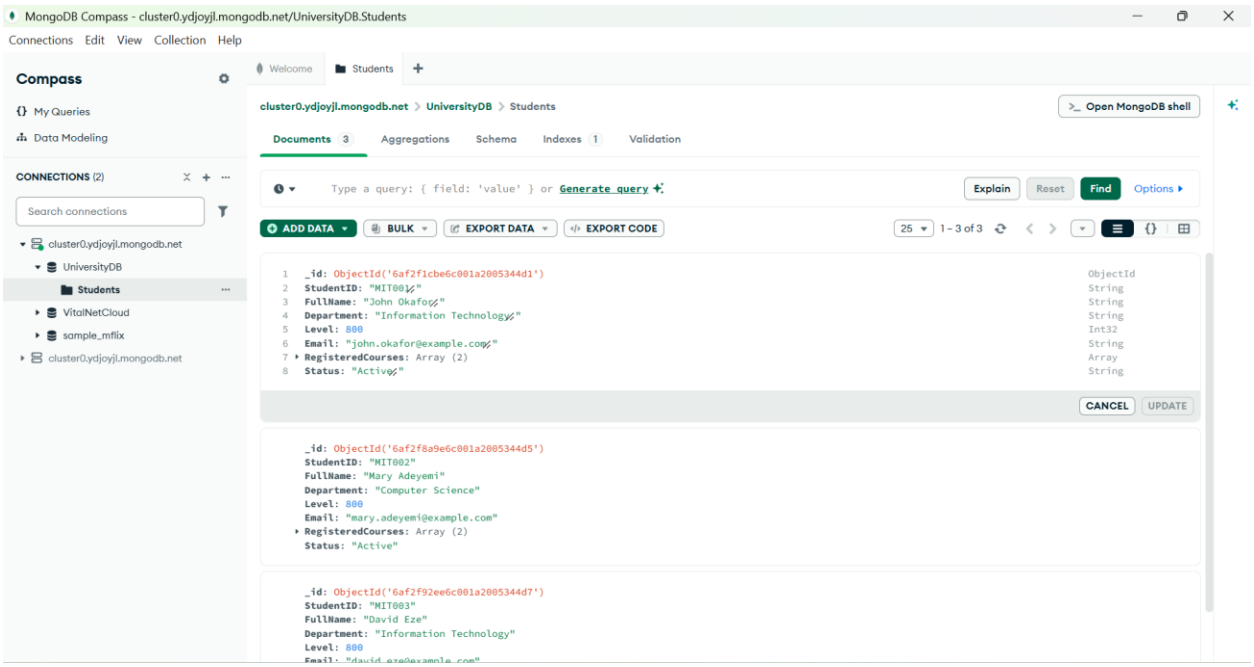

MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 4
Aggregations
Schema
Indexes 1
Validation
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
1 - 4 of 4
_id: ObjectId('6af2f1cbc6c801a2005344d1')
StudentID: "MIT001"
FullName: "John Okafor"
Department: "Information Technology"
Level: 800
Email: "john.okafor@example.com"
RegisteredCourses: Array (2)
Status: "Active"
_id: ObjectId('6af2f8a9c6c801a2005344d5')
StudentID: "MIT002"
FullName: "Mary Adeyemi"
Department: "Computer Science"
Level: 800
Email: "mary.adeyemi@example.com"
RegisteredCourses: Array (2)
Status: "Active"
_id: ObjectId('6af2f92ee6c801a2005344d7')
StudentID: "MIT003"
FullName: "David Eze"
Department: "Information Technology"
Level: 800
Email: "david.eze@example.com"
RegisteredCourses: Array (2)
Status: "Active"
_id: ObjectId('6af3130ac6c801a2005344e5')
StudentID: "TEST001"
FullName: "Temporary Test Student"
Department: "Database Testing"
Level: 100
Email: "test.student@example.com"
RegisteredCourses: Array (1)
Status: "Temporary"

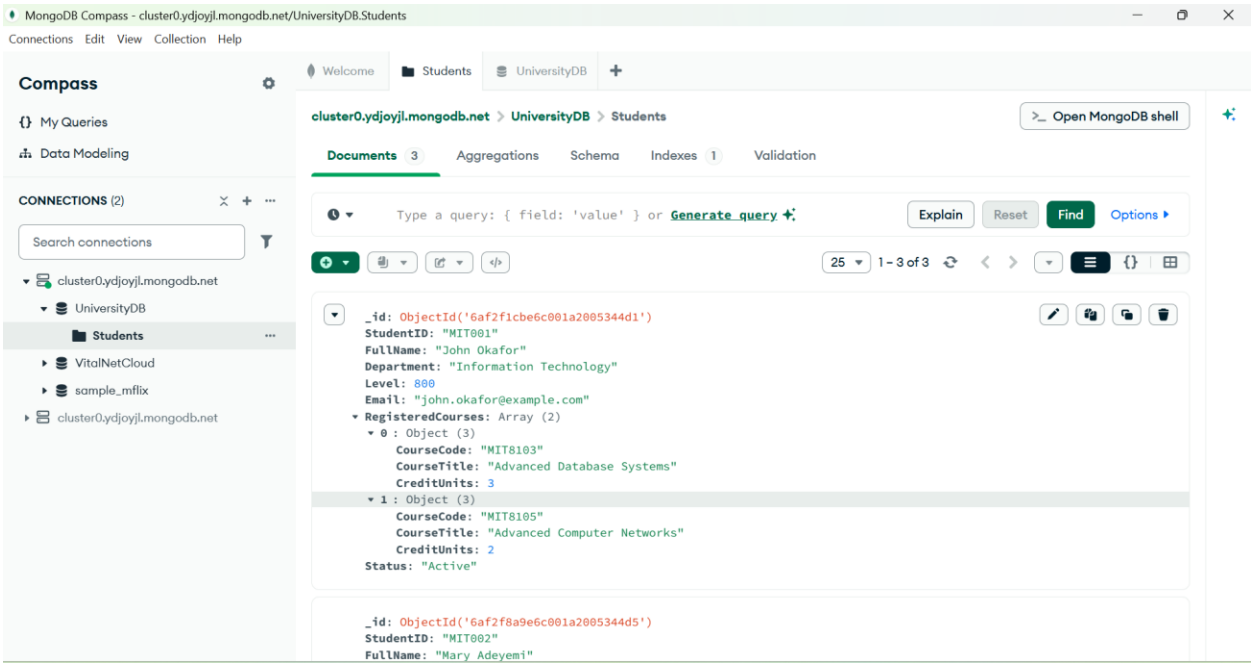

MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 4
Aggregations
Schema
Indexes 1
Validation
{ "StudentID": "TEST001" }
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
Remove document
_id: ObjectId('6af3130ae6c001a2005344e5')
StudentID: "TEST001"
FullName: "Temporary Test Student"
Department: "Database Testing"
Level: 100
Email: "test.student@example.com"
RegisteredCourses: Array (1)
Status: "Temporary"


MongoDB Compass - cluster0.ydjoyjl.mongodb.net/UniversityDB.Students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
cluster0.ydjoyjl.mongodb.net
UniversityDB
Students
VitalNetCloud
sample_mflix
Welcome
My Queries
Data Modeling
Students
UniversityDB
cluster0.ydjoyjl.mongodb.net > UniversityDB > Students
Open MongoDB shell
Documents 4
Aggregations
Schema
Indexes 1
Validation
{ "StudentID": "TEST001" }
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('6af3130ae6c001a2005344e5')
StudentID: "TEST001"
FullName: "Temporary Test Student"
Department: "Database Testing"
Level: 100
Email: "test.student@example.com"
RegisteredCourses: Array (1)
Status: "Temporary"
Doument flagged for deletion.
CANCEL
DELETE



Demonstrating an Embedded Document Model which is performed by the Adding of Course Details inside of John's Student Record.





Explanation of Suitability in Comparison to MySQL

MongoDB is a suitable Database Model for the University Student Management System because it is able to keep data in JSON-like documents, and its flexible. With this, the Database is able to store the Information of a Student, as well as his or her Related Course information in a single document.

In this exercise, every Student's Name, Matriculation Number, Department, Level, Email Address, and Registered Courses, with the Status were stored in the **Students Collection**.

Then, an Array of Embedded Documents is used by the **RegisteredCourses** Field. The Course Code, the Course Title, and the Credit Units are contained in each Embedded Document. With this, every student can Register for different amounts of courses without a fixed number of Course Columns.

In addition, the **Schema is Flexible** as other Student and Course-Related attributes can be brought in without the need to redesign a fixed table structure.

Data Retrieval is Efficient as Students' Information as well as Embedded Course Details can be retrieved together.

Scalability of the Database is ensured as MongoDb provides Horizontal Scaling with Sharding as the application grows.

Cloud Deployment with MongoDB Atlas gives us a Managed Cloud Database Environment.

Availability is ensured because MongoDB replica sets are able to keep multiple copies of Data and Support Failover.

Finally, both MongoDB and MySQL perform complementary purposes in this exercise. For structured records like Students, Courses, Departments, Exam Results, and Lecturers, MySQL is applicable as Relationships and Referential Integrity are vital.

While for storing flexible Student Documents and Embedded Course Information, MongoDB is appropriate and overall, both database models show how Relational and NoSQL implementations support application requirements.